

Feature Perturbation vs. Selection Shift: Post-hoc Calibration of Classical Tabular Models

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Abstract

Temperature scaling and related post-hoc maps are often treated as a cheap fix for miscalibrated probabilities. We measure whether that fix transfers under two different test-time changes, for logistic regression, random forests, and histogram gradient boosting on small tabular datasets (no deep nets, no API models). *Feature mean-shift with frozen labels*—a perturbation of X that does **not** preserve $P(Y | X)$ —raises ECE: at strength 1.5, uncalibrated $\Delta\text{ECE} = \text{ECE}_{\text{pert}} - \text{ECE}_{\text{iid}}$ is positive in 51/60 seed-level cells (mean 0.113, cell-to-cell sd 0.149). That pool mixes sklearn sets (mean 0.042, 25/30 positive) with synthetics (mean 0.184, 26/30 positive) where accuracy also falls. Averaging seeds first, all 12/12 uncalibrated dataset $\times$ model means are positive (a sign count; models that share a dataset are not independent). After i.i.d. temperature, isotonic, or histogram maps, pooled ΔECE remains positive. The same maps can help or hurt relative to doing nothing. By contrast, *selection* on X that keeps the original (X, y) pairs—closer to covariate shift, but a milder change in accuracy—yields smaller ΔECE (quantile-slice cell-mean 0.016 on 6 dataset $\times$ model cells from 2 datasets). Unweighted split-conformal coverage falls under feature perturbation. This is an analysis result, not a new calibrator.

1 Introduction

A classifier is *calibrated* when a predicted probability p is correct with frequency p . Post-hoc maps—Platt-style scaling, histogram binning, isotonic regression, and temperature scaling—are the standard way to improve calibration without retraining [1–3]. A separate literature shows that deep-net uncertainty *collapses* under dataset shift [4–6]. Almost all of that evidence is vision-scale neural nets. Tree ensembles remain strong on tabular data [7], and their i.i.d. calibration properties have been known for decades [2]. Within our API-verified set of 47 papers—mostly deep-net calibration and conformal theory, not a systematic search of applied tabular-drift venues—we did not find this exact measurement (sklearn models, i.i.d. post-hoc maps, test-time feature perturbation vs. selection). That is a scoped gap, not a proof of absence.

We treat that as a measurement question. We do not introduce a new calibrator. A negative or mixed answer is an

acceptable outcome.

Primary hypothesis (reframed). I.i.d.-fitted post-hoc maps do not keep ECE from rising under a Gaussian mean-shift of test features with *labels held fixed* (input perturbation / a change in $P(Y | X)$ at the observed x). We do **not** claim that genuine covariate shift—selection on X that preserves pairings—has the same effect; that is tested as a control.

Two other hypotheses were piloted and *not* selected as the paper claim: isotonic overfit vs. temperature as a function of n_{cal} (H2; heterogeneous), and oracle-weighted conformal recovery (H3; the 1-D weight estimator failed). Decision log: `logs/decisions.md`.

2 Related work

Surveys of calibration methods and metrics include [8, 9]. ECE via confidence binning [3, 10] is biased and definition-dependent [11–13]; we therefore also report Brier score [14] and NLL [15] (pooled Δ alongside ECE in Results). Temperature scaling [3] and Dirichlet calibration [16] target neural nets. Unsupervised recalibration under covariate shift exists for neural models (Pampari and Ermon [17]). Domain-drift post-hoc calibration of CNNs is studied by Tomani et al. [6]. Those papers already *predict* our headline outcomes: ECE should rise under shift, and i.i.d.-fitted post-hoc maps should not be a complete fix. A surprising result here would have been that logistic regression, random forests, and histogram boosting *stay* calibrated under a mean-shift of used covariates *with labels frozen*. Selection that preserves (X, y) pairs is a different mechanism. We reuse the questions, not the model class.

Conformal prediction has finite-sample coverage under exchangeability [18–20] and requires density-ratio weights under covariate shift [21]. We use unweighted split conformal as a secondary diagnostic, not as a proposed fix.

3 Methods

Models. Scikit-learn logistic regression
(with standard scaling), random forests, and

HistGradientBoostingClassifier, with hyperparameters as in `src/calibshift/models.py`. All expose `predict_proba`.

Calibrators. Fitted on a held-out calibration split of the *source* distribution, then frozen. `none`: identity. `temperature`: one scalar T on $\log p$, fitted by NLL (Guo-style, using log-probabilities so trees without logits still apply). `isotonic`: sklearn isotonic regression on $p(y=1)$ (binary) or on confidence (multiclass). `histogram`: equal-width binning of the same 1-D score.

Shifts. *Feature perturbation*: add $s \cdot \hat{\sigma}_j$ to listed test columns, **keeping the original labels**. This moves x without moving y , so it is *not* covariate shift ($P(Y | X)$ is not invariant). $\hat{\sigma}$ is the test-split feature sd. *Selection*: quantile slice and importance resampling of existing (X, y) pairs (closer to covariate shift). Trailing-column perturbation is a control.

Metrics. Equal-width confidence ECE with 15 bins (from the `exp04` config), Brier, NLL, accuracy, and split conformal coverage with the inverse-probability (LAC) score $1 - p_y$ at $\alpha = 0.1$ (not APS).

Protocol. Train/calibration/test partitions follow `calibshift.runner` (source-distribution splits; calibration data never includes the test set), stratified, with the 5 seeds listed in the `exp04` config. Results are written only by `calibshift.io.write_result`. `paper/numbers.tex` is generated from those JSON files.

4 Experiments

Datasets. Offline-only: sklearn `breast_cancer` (binary) and `wine` (3-class); two `make_classification` sets (binary and 3-class). No OpenML, no paid APIs.

Main grid (`exp04_main_h1`). 240 cells, 0 failed, 23.0 seconds on CPU. This is the source of all headline numbers.

Ablations. `exp05`: quantile slice and importance resampling. `exp06`: calibration-set size (killed primary H2). `exp07`: shift trailing columns.

Sanity. I.i.d. logistic regression on breast-cancer: accuracy 0.979, ECE 0.032 ± 0.010 (linear models on this set are typically $\gtrsim 0.95$ accurate). Wine random forests are overconfident i.i.d. (ECE 0.147 ± 0.020), matching the classical tree-calibration picture [2]. Split conformal coverage at $s=0$ is near the nominal $1 - \alpha = 0.9$ for $\alpha=0.1$. We do *not* compare ECE magnitudes to ImageNet temperature scaling [3].

Table 1: Seed-pooled Δ ECE at feature-perturbation $s=1.5$. “std” is heterogeneity, not se of the mean. The $n = 60$ rows are 4 datasets $\times$ 3 models $\times$ 5 seeds (pseudo-replicated). Sklearn/synthetic rows are uncalibrated subsets. One-sided Wilcoxon p -values on this pool are exploratory and live in `results/stats_tests.json` (22 tests, unadjusted).

Calibrator	mean Δ ECE	std	#pos/neg
none	0.113	0.149	51/9
none (sklearn)	0.042	0.039	25/5
none (synthetic)	0.184	0.183	26/4
temperature	0.124	0.150	56/4
isotonic	0.141	0.151	53/7
histogram	0.141	0.151	57/3

5 Results

5.1 Feature perturbation usually raises ECE

Figure 1 plots shifted ECE against perturbation strength s . Figure 2 shows mean Δ ECE at $s=1.5$. Pooled over 60 dataset $\times$ model $\times$ seed cells, uncalibrated mean Δ ECE = 0.113 with cell-to-cell sd 0.149 (51 positive, 9 negative). The same cells have mean Δ Brier 0.257, mean Δ NLL 0.792, and mean accuracy change -0.156. Sklearn-only cells (breast-cancer and wine) have mean Δ ECE = 0.042 (25/30 positive); synthetics have mean 0.184 (26/30 positive). After i.i.d. temperature scaling the pooled mean Δ ECE is 0.124 (sd 0.150). These sd figures describe heterogeneity, not a confidence interval for the mean. Averaging seeds first, all 12 of 12 *uncalibrated* dataset $\times$ model means are positive. An exploratory one-sided Wilcoxon on those 12 means gives $p=2.4e-04$, but models that share a dataset are not independent; a dataset-level one-sided sign test on 4 dataset means is $p=0.0625$. Breast-cancer histogram boosting is the smallest cell (mean 0.002, seed sd 0.009). Synthetic logistic regression is the largest (mean 0.292, sd 0.214) and coincides with a large accuracy drop. Table 1 records seed-pooled sign counts, not confirmatory p -values; `results/stats_tests.json` contains 22 related one-sided tests with no multiplicity correction.

5.2 Post-hoc maps are not a reliable remedy

Figure 3 plots $\text{ECE}_{\text{shift}}(\text{cal}) - \text{ECE}_{\text{shift}}(\text{none})$. Wine random forests *benefit* from isotonic (-0.123 ECE; shifted ECE $0.175 \rightarrow 0.052$), because the uncalibrated forest was already badly calibrated. Temperature scaling on the same cell reaches shifted ECE 0.068. Breast-cancer logistic regression is *hurt* by isotonic (0.074). We therefore do not rank calibrators under shift.

5.3 Coverage under perturbation

Unweighted split conformal (logreg, none, LAC score $1 - p_y$) coverage at $\alpha = 0.1$ (seed mean $\pm$ sd): breast-cancer 0.913 ± 0.015 ($s=0$), 0.778 ± 0.045 ($s=1.5$), 0.608 ($s=2.5$);

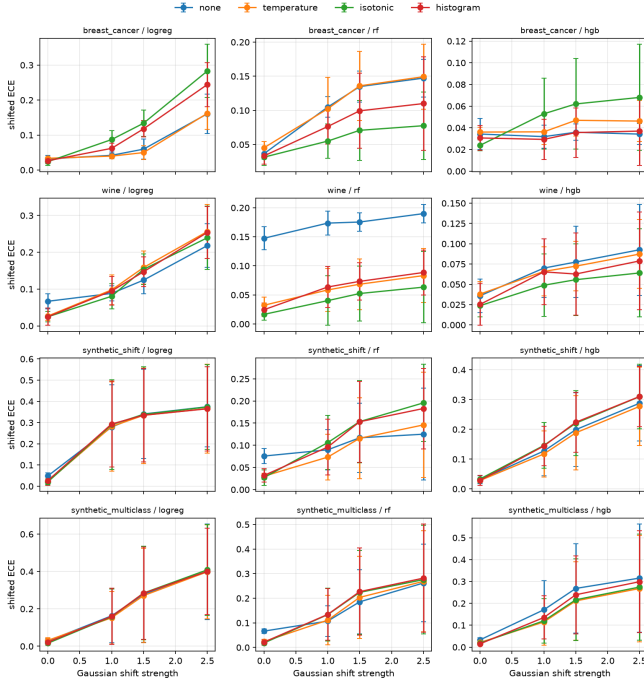


Figure 1: Shifted ECE vs. Gaussian feature-perturbation strength (mean $\pm$ seed sd). $s=0$ is i.i.d. Labels are frozen; this is not covariate shift. Source: results/exp04_main.h1.json.

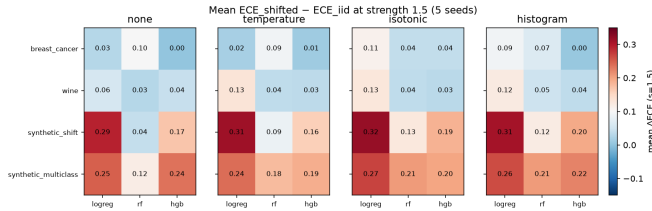


Figure 2: Mean Δ ECE at $s=1.5$. Positive (red) means the shifted test is *worse* calibrated than the i.i.d. test.

synthetic binary 0.634 ± 0.190 at $s=1.5$ (Figure 4). Exchangeability is broken by construction. A 1-D “oracle” density ratio on feature 0 did *not* restore coverage in the H3 pilot (killed).

5.4 Selection shift is a different, weaker effect

Quantile slicing and importance resampling keep original (X, y) pairs, so $P(Y | X)$ is preserved and the mechanism is closer to covariate shift. This ablation uses 6 dataset $\times$ model cells on 2 datasets (breast-cancer and synthetic_shift; wine and the multiclass synthetic are absent). Mean Δ ECE (none) is 0.016 (sd 0.006) for quantile slice and 0.011 (sd 0.031) for importance resampling. On the *same* 6 cells, Gaussian perturbation at $s=1.5$ has cell-mean Δ ECE 0.105 (sd 0.109). Quantile slice also moves accuracy less (cell-mean Δ acc -0.016 vs. matched Gaussian -0.128), so the gap is not identified as “preserving $P(Y | X)$ ” alone. Importance resampling can even *decrease* ECE on breast-cancer. This control

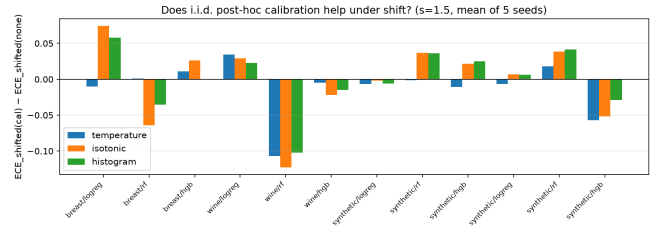


Figure 3: Does i.i.d. calibration help on the shifted test? Negative bars mean the calibrator beats *none* under shift.

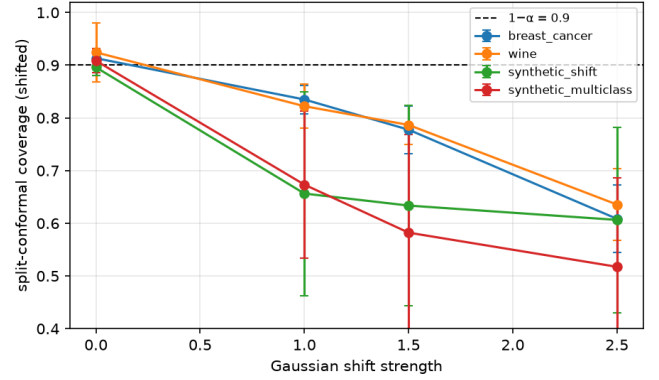


Figure 4: Unweighted split-conformal coverage vs. feature perturbation (logreg, no calibrator, LAC score). Dashed line: $1 - \alpha = 0.9$.

does not support a claim that i.i.d. post-hoc calibration fails under covariate shift in general. Trailing-column perturbation on synthetic_shift yields RF Δ ECE = 0.009 vs. 0.041 when perturbing the first four columns.

6 Limitations

- **The headline mechanism is not covariate shift.** Frozen-label feature mean-shift changes $P(Y | X)$ at the observed x (input corruption). Selection controls that preserve pairings show small Δ ECE. Title and abstract are scoped accordingly.
- **Shift is synthetic** and, for feature perturbation, entangled with accuracy drop (synthetic logreg $0.932 \rightarrow 0.620$ at $s=1.5$).
- **Small tabular data only.** Wine test splits have 45 points and 44 calibration points with 15 ECE bins; those cells are noisy. No ImageNet, no LMs, no naturally occurring drift (no Folktables/WILDS).
- **Multiclass isotonic/histogram** map confidence, not a full Dirichlet map [16]; argmax can flip after renormalization.
- **Temperature on probabilities.** Trees have no logits; RF 0/1 probabilities saturate after clipping. This is not Guo logit temperature.

- **H3 weights were misspecified.** A 1-D histogram ratio is not $p_{\text{test}}(x)/p_{\text{cal}}(x)$ for a 4-D perturbation.
- **Pooling and tests.** Seed-level $n = 60$ Wilcoxon is pseudo-replicated and one-sided. The $n = 12$ seed-averaged uncalibrated means are a sign count, not independent replicates. A dataset-level one-sided sign test has $n = 4$ and $p=0.0625$. 22 tests, unadjusted. Selection comparisons use 6 cells on 2 datasets and are not strength-matched to $s=1.5$ perturbation.
- **Search scope.** Novelty is relative to `paper/verified.bib` (47 API-verified papers).

7 Conclusion

For these sklearn models, i.i.d. post-hoc calibration is not a dependable answer to *test-time feature perturbation with frozen labels*: ECE usually rises and conformal coverage falls, while the maps can make perturbed ECE better or worse than doing nothing. The same experiments **do not** show a comparable failure under selection on X . The HGB exception and the killed H2/H3 hypotheses are part of the result.

AI-generation disclosure

This repository and draft were produced by a coding agent executing the user’s GOAL.md protocol (literature verification via arXiv and Semantic Scholar APIs, experiments on CPU, no hand-edited `results/*.json`). Every numeric macro in `paper/numbers.tex` is generated from those files. Citations are restricted to `paper/verified.bib`. A human should still read the limitations and re-run `make fresh-clone-test` before treating the PDF as camera-ready.

References

- [1] B. Zadrozny and C. Elkan. Transforming classifier scores into accurate multiclass probability estimates. 2002. doi: 10.1145/775047.775151. URL <https://doi.org/10.1145/775047.775151>.
- [2] Alexandru Niculescu-Mizil and R. Caruana. Predicting good probabilities with supervised learning. 2005. doi: 10.1145/1102351.1102430. URL <https://doi.org/10.1145/1102351.1102430>.
- [3] Chuan Guo, Geoff Pleiss, Yu Sun, and Kilian Q. Weinberger. On Calibration of Modern Neural Networks. International Conference on Machine Learning, 2017. URL <https://arxiv.org/abs/1706.04599>.
- [4] Yaniv Ovadia, Emily Fertig, Jie Jessie Ren, Zachary Nado, D. Sculley, Sebastian Nowozin, Joshua V. Dillon, Balaji Lakshminarayanan, and Jasper Snoek. Can You Trust Your Model’s Uncertainty? Evaluating Predictive Uncertainty Under Dataset Shift. Neural Information Processing Systems, 2019. URL <https://arxiv.org/abs/1906.02530>.
- [5] M. Minderer, J. Djolonga, Rob Romijnders, F. Hubis, Xiaohua Zhai, N. Houlsby, Dustin Tran, and M. Lučić. Revisiting the Calibration of Modern Neural Networks. Neural Information Processing Systems, 2021. URL <https://arxiv.org/abs/2106.07998>.
- [6] Christian Tomani, Sebastian Gruber, Muhammed Ebrar Erdem, Daniel Cremers, and Florian Buettner. Post-hoc Uncertainty Calibration for Domain Drift Scenarios, 2020. URL <https://arxiv.org/abs/2012.10988>.
- [7] Léo Grinsztajn, Edouard Oyallon, and Gaël Varoquaux. Why do tree-based models still outperform deep learning on tabular data?, 2022. URL <https://arxiv.org/abs/2207.08815>.
- [8] T. S. Filho, Hao Song, Miquel Perello-Nieto, R. Santos-Rodríguez, Meelis Kull, and Peter A. Flach. Classifier calibration: a survey on how to assess and improve predicted class probabilities. Machine Learning, 2021. URL <https://arxiv.org/abs/2112.10327>.
- [9] Cheng Wang. Calibration in Deep Learning: A Survey of the State-of-the-Art. arXiv.org, 2023. URL <https://arxiv.org/abs/2308.01222>.
- [10] Mahdi Pakdaman Naeini, G. Cooper, and M. Hauskrecht. Obtaining Well Calibrated Probabilities Using Bayesian Binning. 2015. doi: 10.1609/aaai.v29i1.9602. URL <https://doi.org/10.1609/aaai.v29i1.9602>.
- [11] J. Nixon, Michael W. Dusenberry, Linchuan Zhang, Ghassen Jerfel, and Dustin Tran. Measuring Calibration in Deep Learning. CVPR Workshops, 2019. URL <https://arxiv.org/abs/1904.01685>.
- [12] Ananya Kumar, Percy Liang, and Tengyu Ma. Verified Uncertainty Calibration. Neural Information Processing Systems, 2019. URL <https://arxiv.org/abs/1909.10155>.
- [13] Juozas Vaicenavicius, David Widmann, Carl R. Andersson, F. Lindsten, Jacob Roll, and Thomas Bo Schön. Evaluating model calibration in classification. International Conference on Artificial Intelligence and Statistics, 2019. URL <https://arxiv.org/abs/1902.06977>.
- [14] G. Brier. VERIFICATION OF FORECASTS EXPRESSED IN TERMS OF PROBABILITY. 1950. doi: 10.1175/1520-0493(1950)078<0001:VOFEIT>2.0.CO;2. URL [https://doi.org/10.1175/1520-0493\(1950\)078<0001:VOFEIT>2.0.CO;2](https://doi.org/10.1175/1520-0493(1950)078<0001:VOFEIT>2.0.CO;2).

- [15] T. Gneiting and A. Raftery. Strictly Proper Scoring Rules, Prediction, and Estimation. 2007. doi: 10.1198/016214506000001437. URL <https://doi.org/10.1198/016214506000001437>.
- [16] Meelis Kull, Miquel Perello-Nieto, Markus Kängsepp, T. S. Filho, Hao Song, and Peter A. Flach. Beyond temperature scaling: Obtaining well-calibrated multi-class probabilities with Dirichlet calibration. *Neural Information Processing Systems*, 2019. URL <https://arxiv.org/abs/1910.12656>.
- [17] Anusri Pampari and S. Ermon. Unsupervised Calibration under Covariate Shift. *arXiv.org*, 2020. URL <https://arxiv.org/abs/2006.16405>.
- [18] Vladimir Vovk, Alex Gammerman, and Glenn Shafer. *Algorithmic Learning in a Random World*. 2005. doi: 10.1007/b106715. URL <https://doi.org/10.1007/b106715>.
- [19] Jing Lei, M. G'Sell, A. Rinaldo, R. Tibshirani, and L. Wasserman. Distribution-Free Predictive Inference for Regression. *Journal of the American Statistical Association*, 2016. URL <https://arxiv.org/abs/1604.04173>.
- [20] Anastasios Nikolas Angelopoulos and Stephen Bates. A Gentle Introduction to Conformal Prediction and Distribution-Free Uncertainty Quantification. *arXiv.org*, 2021. URL <https://arxiv.org/abs/2107.07511>.
- [21] R. Tibshirani, R. Barber, E. Candès, and Aaditya Ramdas. Conformal Prediction Under Covariate Shift. *Neural Information Processing Systems*, 2019. URL <https://arxiv.org/abs/1904.06019>.